

EuroEats

Data & Analytics Case Study

Prepared by: EuroEats Operations & Data Team | For: Data Analyst / Data Scientist Candidate
Assignment

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Company Overview

EuroEats is a food-delivery marketplace founded in 2023 and headquartered in Amsterdam, the Netherlands. What started as a single-city pilot connecting independent restaurants with local couriers has grown, in just under three years, into a platform operating across 15 European countries. The company was built on a simple observation: most delivery platforms optimized for scale at the expense of the small, independent restaurants and drivers who actually make the experience work. EuroEats set out to build a leaner, fairer alternative — one with lower commission rates for restaurant partners and more transparent, predictable pay for drivers.

Our story

EuroEats launched in Amsterdam in early 2023 with a handful of restaurant partners and a small courier fleet. After a strong first year, the company expanded into Germany, France, and the Benelux region, before completing a Series A funding round that fueled expansion into Southern and Northern Europe. Today, EuroEats is one of the fastest-growing independent food-delivery platforms in the region, with a reputation for reliability in markets larger competitors have deprioritized.

Leadership

- **Sofia Álvarez — Co-Founder & CEO.** Previously led regional operations at a European logistics company before starting EuroEats to fix what she saw as a broken deal for restaurants and drivers.
- **Jonas Bergström — Co-Founder & CTO.** Former mapping-and-routing engineer who built EuroEats' early dispatch and ETA systems, and still owns the platform's technical roadmap today.

Quick facts

- Founded: 2023, Amsterdam, Netherlands
- Markets: 15 countries across Europe
- Focus: independent restaurants, fair driver pay, reliable delivery estimates
- Stage: post-Series A, scaling operations across existing markets

Company & Data Overview

EuroEats is a food-delivery marketplace operating in 15 European countries, connecting customers, restaurant partners, and independent delivery drivers. The data team has extracted the following tables from the production database for this assignment. Column-level definitions are provided in the accompanying data dictionary; the diagram on the next page shows how the tables relate to one another.

Tables provided

- **customers** — everyone who has signed up to order on the platform.
- **restaurants** — partner restaurants listed on the platform.
- **menu_items** — the dishes and drinks each restaurant sells.
- **drivers** — the delivery workforce.
- **orders** — every order placed, with pricing and status.
- **order_items** — the individual line items within each order.
- **deliveries** — trip-level detail for orders that were dispatched to a driver.
- **payments** — payment transactions tied to orders.
- **reviews** — customer feedback left after an order.

Use these tables (and only these tables) to answer every question in this pack.

Entity Relationship Diagram

The 9 tables are organized around **orders** as the central record, with optional one-to-one extensions (deliveries, payments, reviews) and standard one-to-many lookup tables (customers, restaurants, menu_items, drivers, order_items).

EuroEats — Entity Relationship Diagram

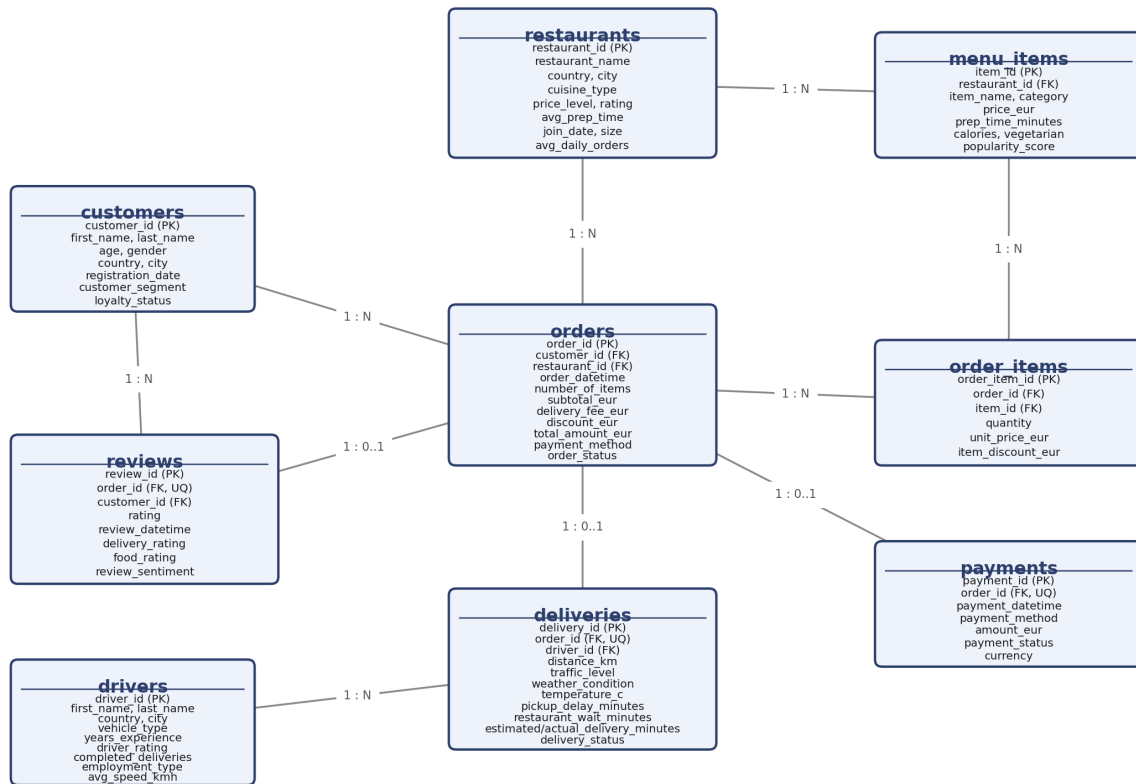


Figure 1 — EuroEats relational schema.

Business Problem Statement

Over the last two quarters, EuroEats's customer support team has seen a steady rise in complaints about delivery estimates: the time shown to a customer at checkout frequently doesn't match how long the order actually takes to arrive. Leadership is concerned this is quietly hurting repeat-order rates and driving up support costs, but no one has yet quantified how big the gap is, where it is worst, or what's driving it.

The ask

The Operations and Product teams have asked the Data team to (1) investigate what is really driving long or unpredictable deliveries across EuroEats's markets, and (2) produce a model that can predict how long a new order will actually take to deliver, before it is dispatched, so the checkout screen can show customers a number they can trust. This is expected to be a **regression** problem, since the outcome the business cares about — delivery duration — is a continuous number of minutes, not a category.

What you're asked to deliver

- A short written analysis of what drives delivery time and where performance is weakest (see Business Analysis Questions).
- A set of SQL queries answering the operational and commercial questions the business has raised (see SQL Assignment).
- A predictive model for delivery duration, along with an honest assessment of how accurate it is and where it still falls short.
- A Power BI dashboard the Operations and Leadership teams can use on an ongoing basis (see Dashboard Requirements).

How this will be evaluated

We're less interested in a single accuracy number than in whether your analysis would actually change a decision: could Operations use your findings to fix something, and would the model be trustworthy enough for Product to put in front of customers? Come prepared to explain your reasoning, not just your results.

SQL Assignment

The questions below are grouped by the business area they come from, not by difficulty. Write a single SQL query for each one. Assume you're answering a real stakeholder, not a grader — your query should return exactly what they asked for.

Customer Insights & Loyalty

1. Which customer segments (Budget, Regular, Premium) place the most orders, and how does that mix differ by country?
2. Who are EuroEats's 10 highest lifetime-spend customers?
3. Which customers have had orders delivered but have never left a review?
4. Rank all customers by total spend and group them into spending tiers, so Marketing can target the top tier with a loyalty offer.
5. Which customers haven't placed an order in the 90 days before the most recent order on the platform — good candidates for a win-back campaign?
6. For each customer, show how their total spend has built up over time, so the loyalty team can see who is close to the next reward tier.

Restaurant & Menu Performance

7. Which restaurants bring in the most revenue, and does the answer change by country?
8. What are the 10 most frequently ordered menu items on the platform?
9. Which restaurants have the strongest customer ratings, and how many reviews is that rating actually based on?
10. For each restaurant, what are its three best-selling menu items? Restaurant partners keep asking us this.
11. Which cuisine types offer the highest share of vegetarian dishes, and do those cuisines also see higher order volume?
12. Which restaurants take noticeably longer to prepare food than the platform average, ranked within their own country, so our partnerships team knows who to talk to first?

Revenue, Orders & Promotions

13. What does total revenue look like by country and by month, and is it trending up or down?
14. What is the month-over-month revenue growth rate for the platform?
15. What share of orders use a discount, and does typical discount size differ by country?
16. Which payment methods fail most often, and is it worth flagging any of them to the payments team?
17. How much order value has EuroEats lost to cancellations and refunds, broken down by country?
18. For every country, which single restaurant generates the most revenue — we want a shortlist for a 'featured partner' promotion.

Delivery & Driver Operations

19. How does average delivery time change across different traffic and weather conditions?

- 20.** What share of deliveries come in late, and does that cluster around particular traffic conditions?
- 21.** Which drivers combine a strong rating with high delivery volume — our candidates for a 'driver of the month' recognition program?
- 22.** Does delivery performance differ meaningfully by vehicle type, and should that change how we recruit drivers in any particular city?
- 23.** What is the typical (not just average) delivery time under each traffic condition? We want a realistic number for the checkout ETA, not one skewed by outliers.
- 24.** Are there cases of different customers, in the same city, ordering from the same restaurant on the same day? Ops wants to know if batching these deliveries is worth exploring.

Strategic & Cross-Functional Questions

- 25.** Which countries have the fastest and the slowest average delivery times, and what in the data might explain that gap?
- 26.** Do bigger orders tend to carry bigger discounts, and could that be quietly eating into our margin?
- 27.** Which restaurant-and-city combinations produce delayed deliveries most often, and should be first in line for an operational review?
- 28.** Is there any real relationship between a restaurant's rating and how much business it does, or are they unrelated?
- 29.** Do full-time, part-time, and freelance drivers actually perform differently, or is that a myth?
- 30.** Build one view that connects an order through to its delivery and its review, so Leadership can see operational performance and customer satisfaction side by side in a single quarterly report.

Business Analysis Questions

These are the questions Operations and Product actually want answered. How you get there (queries, spreadsheets, notebooks, charts) is up to you — we care about the answer and how well you can defend it.

1. What actually drives delivery time the most, and how big is each effect in practice?
2. Are certain countries or cities consistently slower to deliver than others, and what might explain it?
3. Does bad weather slow deliveries down enough to justify a weather-based delivery surcharge?
4. Which cuisines bring in the most revenue versus the most orders — and is there a mismatch worth investigating?
5. Are Premium customers actually more valuable to the business, or do they just order more often at a similar order value?
6. Do higher-rated restaurants actually get more business, or is rating disconnected from order volume?
7. Which vehicle types work best for short-distance deliveries versus long-distance ones?
8. Do discounted orders meaningfully increase how much customers buy, or are we mostly discounting orders that would have happened anyway?
9. What is the real financial cost of cancelled and refunded orders, and does it vary a lot by country?
10. Does driver experience actually translate into faster or better-rated deliveries?
11. Which restaurants consistently run slower than their own stated prep time would suggest?
12. How much of the swing in delivery time comes down to time-of-day and day-of-week congestion, versus other factors?
13. Do customers who get a slow delivery leave worse reviews, and does that pattern hold across every country?
14. Which countries or cities look like the best growth opportunity, given how many customers we already have there?
15. Is there a seasonal pattern to revenue or delivery performance that the business should plan around?
16. Do bigger orders take proportionally longer to prepare and deliver, and does that hold across restaurants of every size?
17. How reliable are our delivery-time estimates today, and in which countries is the gap between estimated and actual time worst?
18. What share of revenue comes from repeat customers versus first/one-time customers, and how does that split by segment?
19. Do freelance drivers perform differently from full-time drivers in terms of speed and ratings?
20. Based on what customers say in reviews and how they rate their deliveries, where are the biggest pain points in their experience today?

Power BI Dashboard Requirements

Operations and Leadership have asked for a 3-page Power BI report they can check on a recurring basis, built on the tables above. All three pages should share a common country / date-range / cuisine filter panel.

Page 1 — Executive Overview

"I want to open this page and know within 10 seconds whether the business had a good month." — Head of Operations

Should include:

- Headline numbers: total orders, total revenue, average order value, average delivery time, on-time delivery rate
- Revenue and order volume trend over time
- Revenue and order volume by country
- Revenue split by cuisine type
- Order outcomes: delivered vs. cancelled vs. refunded
- A shortlist of the top-performing restaurants

Filters: Country; Date range; Customer segment; Cuisine type

Page 2 — Delivery & Driver Performance

"I need to know where deliveries are breaking down — which city, which conditions, which drivers — so I can actually fix it." — Operations Manager

Should include:

- Headline numbers: average delivery time, % delayed deliveries, average distance, average driver rating
- Delivery time under different traffic and weather conditions
- Delivery time versus distance, split out by vehicle type
- Delivery time by vehicle type
- Delivery time by hour of day and day of week, to spot peak-hour bottlenecks
- A driver leaderboard: volume, rating, and speed, filterable by vehicle type

Filters: Country / city; Traffic level; Weather condition; Vehicle type; Date range

Page 3 — Customer & Restaurant Health

"I want to see who our best customers and best restaurant partners are, and whether people are actually happy with their experience." — Head of Product

Should include:

- Headline numbers: total customers, repeat-order rate, average rating given, average restaurant rating
- Customer segment mix by country

- Loyalty tier breakdown (Bronze/Silver/Gold/Platinum)
- Top restaurants by rating and by revenue
- Review sentiment breakdown
- Restaurant rating versus order volume, to spot under-recognized partners

Filters: Country; Customer segment; Loyalty status; Cuisine type; Date range